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2022-02-07 19:20 ChanServ changed the topic of #rust-embedded to: Welcome to the Rust Embedded IRC channel! Bridged to #rust-embedded:matrix.org and logged at <https://libera.irclog.whitequark.org/rust-embedded>, code of conduct at <https://www.rust-lang.org/conduct.html>

04:56 <korken89[m]> I've just started a new FPGA project based on ECP5 85k. :) nothing fancy, just a sampling rig to connect 64 DUTs with 3 important signals and one defmt logging UART so it will do RLE encoding of the signals and decode the UART streams, to finally streaming it all over USB 3.

04:58 <korken89[m]> So one can test all the DUTs (RF devices) in parallel and time synched to make sure they do TDMA correctly

05:20 <korken89[m]> Doodling the gateway to test the sub components and I run Amaranth for this

05:21 <korken89[m]> <adamgreig[m]> "ah, I was looking for https://..." <- This one looked interesting though

05:37 <korken89[m]> This reminds me that I need to get adamgreig's `ecpdap` working with the rusty probe again 😊 I did not port the minimal JTAG support there was in the HS probe

05:59 <juliand[m]> Does amaranth have better documentation now?

05:59 <juliand[m]> I remember trying it out a few months ago on the ice40 and it was a pain to set up, some instructions in the guide didn't work and not much of the "language's" functionality had been documented. Like how you interact with these python modules/components and how to get the toolchain set up etc. Also lacking a couple of more sophisticated examples iirc.

05:59 <juliand[m]> Really liked the approach though, and it seems to have some commercial backing. So that's very promising.

06:01 <korken89[m]> Not sure tbh, when I upgraded from nmigen it was quite straight forward and I generally spelunk the codebase when I want to know something :/

06:07 <juliand[m]> Hmm, that's what I did as well to figure some things out, but that's not a good experience to get you started. However, I am far from an FPGA expert and my experience with VHDL and Verilog is mostly from university... maybe that was my main issue there 😊

06:23 <korken89[m]> When I started there was a looot of good tutorials for nMigen, so I have just extrapolated from there :)

06:24 <korken89[m]> E.g. there was a guy writing an entire RISC-V from scratch with nMigen, that was really insightful

06:26 <korken89[m]> Also they are really helpful in my experience, just ask in #amaranth-lang:matrix.org and you'll get good feedback :)

06:27 <korken89[m]> Last time I was really using nMigen there was a IRQ chat, maybe this exists still as well

06:52 emerent_ is now known as emerent

06:57 <juliand[m]> Cool, thank you! Will give it a try soon :)

07:07 <korken89[m]> No problem!

07:16 <korken89[m]> * parallel and measure time synched

07:45 <korken89[m]> Also if you didn't know already, this is quite a nice collection so one does not need to install everything <https://github.com/YosysHQ/oss-cad-suite-build>

07:46 <korken89[m]> * to install/build everything

07:50 <juliand[m]> Yes, that I found already, works like a charm! (Unless you install the arm64 instead of amd64 binaries on an Intel Laptop and wonder why the instruction set is supposedly wrong, lol)

08:04 <korken89[m]> xD

10:32 <thejpster[m]> One final ping on <https://github.com/rust-embedded-community/embedded-sdmmc-rs/pull/100> before I just merge it myself.

11:35 <adamgreig[m]> juliand: the docs are being worked on, there's been a lot of renewed development work in amaranth in the last couple of months

12:22 <korken89[m]> <adamgreig[m]> "juliand: the docs are being..." <- Because of Glasgow coming? :D

12:23 <adamgreig[m]> I think it's more to do with Catherine getting a job that partially involves working on amaranth